

XGBoost Model Performance Summary

Automated Report

2025-11-28

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1 Overview

This report summarizes the performance of the current XGBoost regression model trained on the formatted feature matrix with the outcome scaled to the range [-1, 1].

Table 1: Model vs. Shuffled Baseline Metrics

Metric	Value
rmse_test	0.28140
rmse_random_mean	0.30660
rmse_random_sd	0.04669
improvement	0.02526
z_score	0.54110
approx_p_value	0.29420
mae_test	0.23080
r2_test	-0.25720
spearman_test	0.14290

1.1 Interpretation

Compared to a random shuffle of the training labels (null baseline), the model achieved:

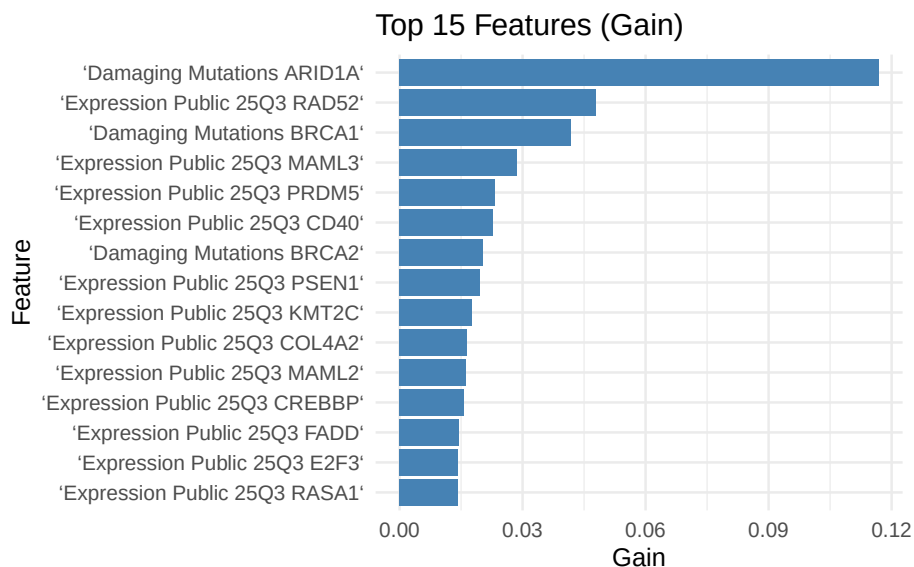
- Test RMSE: **0.2814**
- Mean shuffled RMSE: **0.3066** (SD **0.0467**)
- Absolute improvement (lower RMSE vs. shuffled mean): **0.0253**
- Approximate z-score (improvement / SD of shuffled RMSEs): **0.54**
- Approximate one-sided p-value: **0.294** (probability a random model is at least this good under a normal approximation)

The positive improvement and low RMSE relative to the shuffled baseline indicate the model captures signal beyond chance, though the modest z-score suggests the margin is limited and further optimization or feature engineering could be beneficial.

R-squared is **-0.257** (can be negative for models worse than the mean baseline under scaling). Spearman correlation **0.143** provides a rank-based agreement measure.

1.2 Feature Importance (Gain)

The top features by XGBoost gain demonstrate which predictors most contributed to reducing loss across splits.



A full CSV of importance values is saved at `Results/feature_importance_gain.csv`.

1.3 Diagnostic Figures

1.3.1 Predicted vs. True

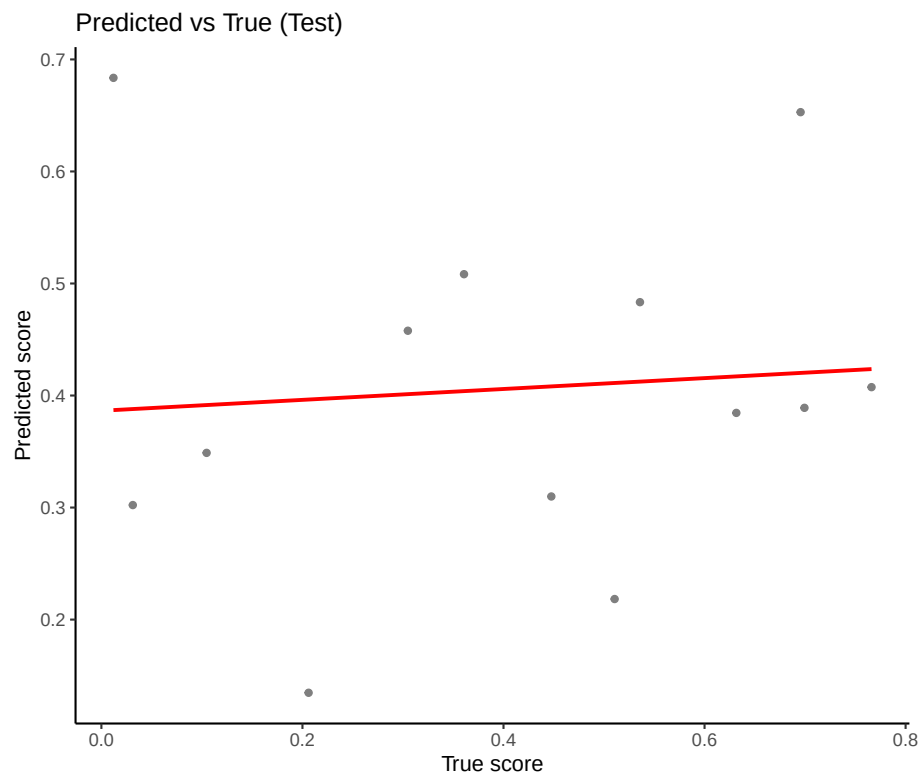


Figure 1: Predicted vs True (Test Set)

1.3.2 Shuffled RMSE Distribution

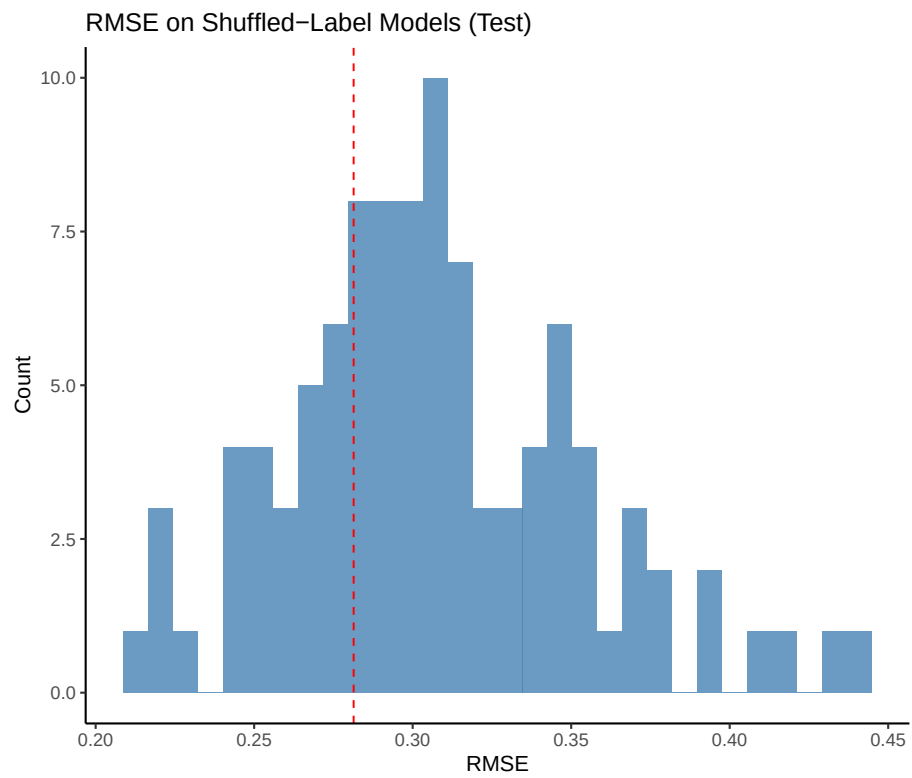


Figure 2: RMSE Distribution for Shuffled-Label Models

1.3.3 Feature Importance Top 30

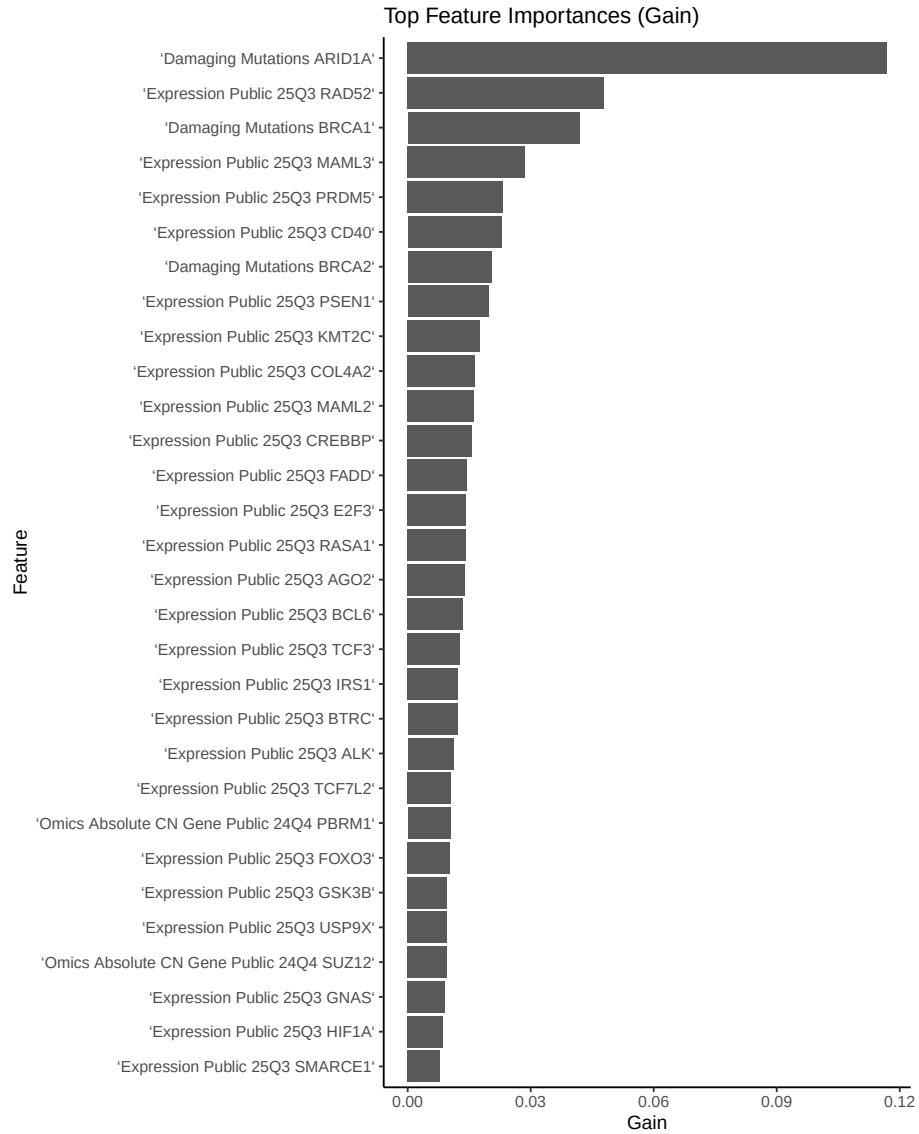


Figure 3: Top 30 Feature Importances (Gain)

1.4 Recommendations

1. Explore narrower tuning around best hyperparameters (e.g., eta and max_depth) to attempt incremental gains.

2. Consider feature grouping or dimensionality reduction (e.g., PCA, pathway aggregation) to amplify signal.
3. Evaluate alternative gradient boosting libraries (LightGBM, CatBoost) for potential performance improvements.
4. Investigate calibration (e.g., isotonic regression) if prediction distribution needs smoothing.

1.5 Session Info

R version 4.3.3 (2024-02-29)

Platform: aarch64-apple-darwin20 (64-bit)

Running under: macOS 26.1

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRlapack.dylib;

locale:

[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8

time zone: Europe/Copenhagen

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] kableExtra_1.4.0 tidyr_1.3.1 ggplot2_3.5.1 dplyr_1.1.4
 [5] data.table_1.14.10 readr_2.1.4 knitr_1.45

loaded via a namespace (and not attached):

[1] gtable_0.3.6 jsonlite_2.0.0 compiler_4.3.3 tidyselect_1.2.1
 [5] xml2_1.3.6 stringr_1.5.1 systemfonts_1.0.5 scales_1.3.0
 [9] yaml_2.3.8 fastmap_1.1.1 R6_2.5.1 labeling_0.4.3
 [13] generics_0.1.3 tibble_3.2.1 munsell_0.5.0 svglite_2.1.3
 [17] pillar_1.9.0 tzdb_0.4.0 rlang_1.1.6 utf8_1.2.4
 [21] stringi_1.8.3 xfun_0.41 viridisLite_0.4.2 cli_3.6.5
 [25] withr_2.5.2 magrittr_2.0.3 digest_0.6.34 grid_4.3.3
 [29] rstudioapi_0.17.1 hms_1.1.3 lifecycle_1.0.4 vctrs_0.6.5
 [33] evaluate_0.23 glue_1.7.0 farver_2.1.1 fansi_1.0.6
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